

Disproof of Conjecture 00000009617: the even shift has a 2-state Fischer cover but Hankel rank 3

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Abstract

Conjecture 00000009617 asserts, among other things, that the Fischer (“Fisher”) minimal state count of a sofic shift equals the rank of the Hankel matrix of its language. We show that this clause is false when rank has its standard meaning, rank over a field. For the even shift X (binary sequences in which every run of 1s between two 0s has even length) the Fischer cover has exactly 2 states, and no presentation of any kind has fewer. But the Hankel matrix $H[u][v] = [uv \in \mathcal{L}(X)]$ has the 3×3 minor $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ with determinant -1 , so its rank is exactly 3 over every field. A second example, the charge-constrained shift X_c , has Fischer count 3, 6 nonempty follower sets and Hankel rank 5. It also defeats the reading of “state count” as the size of the follower-set graph. The decisive facts are formalized and checked in a self-contained Lean 4 project:

- the presented shifts;
- the state counts 2 and ≤ 3 ;
- the language of the even shift for all words;
- the minors with determinant -1 and 1 .

An independent Python script confirms them. Under a Boolean (OR/AND) reading of rank, the clause is not refuted.

1 The conjecture

The statement (English version) reads:

Definition: Sofic shifts are factor images of SFTs, presented by finite graphs with output (hidden Markov models); the Fisher minimal realization is unique. *Conjecture:* The Fisher minimal state count of a sofic shift equals the rank of the Hankel matrix of its language; the rank is computable in polynomial time by a local criterion on words of length $2n + 1$, and the minimal realization graph structure is unique.

The Chinese version says the same. The conjecture is a conjunction of three clauses. We refute the first one:

(C) *For every sofic shift X , the number of states of its Fischer (minimal right-resolving) presentation equals the rank of the Hankel matrix of $\mathcal{L}(X)$.*

A conjunction with a false clause is false, so the conjecture is false. The two later clauses are not needed and we say nothing about them. “Fisher” is a common misspelling of R. Fischer, who introduced the minimal right-resolving presentation [1].

2 Definitions

We follow Lind and Marcus [2]. The alphabet is $\{0, 1\}$. A point is a bi-infinite sequence $x = (x_i)_{i \in \mathbb{Z}}$. A word $w = w_0 \cdots w_{n-1}$ occurs in x at i if $x_{i+j} = w_j$ for all $j < n$. The language $\mathcal{L}(X)$ of a set X of points is the set of words that occur in some point of X .

Labelled graphs and presentations. A labelled graph G on the vertex set $\{0, \dots, n-1\}$ is a set of edges $p \xrightarrow{a} q$ with $a \in \{0, 1\}$. The shift X_G presented by G is the set of label sequences of bi-infinite paths, i.e. of all x for which there are vertices s_i ($i \in \mathbb{Z}$) with an edge $s_i \xrightarrow{x_i} s_{i+1}$ for every i . A shift space X is *sofic* if $X = X_G$ for some finite G ; then G *presents* X . (Parallel edges with the same label do not change X_G , so we may take G to be a set of labelled edges.) G is *right-resolving* if the out-edges of each vertex carry distinct labels.

Fischer cover. Every irreducible sofic shift has a minimal right-resolving presentation, unique up to isomorphism [1], [2, §3.3]. It is called the Fischer cover. It has the fewest vertices among all right-resolving presentations [2, §3.3]. So the “Fisher minimal state count” is

$$\text{Fis}(X) = \min\{n : X \text{ has a right-resolving presentation with } n \text{ vertices}\}.$$

We also consider $\text{Pres}(X) = \min\{n : X \text{ has a presentation with } n \text{ vertices}\}.$

Hankel matrix and rank. The Hankel matrix of a language $\mathcal{L}$ is the infinite 0/1 matrix indexed by pairs of words, $H[u][v] = 1$ if $uv \in \mathcal{L}$ and 0 otherwise. Its row u is the indicator of the follower set $\mathcal{F}(u) = \{v : uv \in \mathcal{L}\}$. The rank of an infinite matrix over a field K is the dimension of its row space. Equivalently, it is the supremum of the ranks of its finite submatrices, or the largest r such that some $r \times r$ minor is nonzero in K . This is the meaning of “Hankel rank” in realization theory, from Kronecker’s theorem on Hankel matrices of rational functions to the Fliess and Carlyle–Paz theorems on rational formal power series and the Hankel rank of hidden Markov processes [3, 4, 5].

3 The even shift and its language

The even shift is

$$X = \{x \in \{0, 1\}^{\mathbb{Z}} : \text{no block } 01^{2k+1}0 \text{ } (k \geq 0) \text{ occurs in } x\}.$$

A block 01^m0 is exactly a run of m ones bounded by 0s on both sides, so X consists of the sequences in which every such run has even length.

Lemma 1. *For every word w : $w \in \mathcal{L}(X)$ if and only if w contains no factor $01^{2k+1}0$.*

Proof. If w occurs in $x \in X$ at i , then a factor $01^{2k+1}0$ of w at position j would occur in x at $i+j$. Conversely, if w has no such factor, let x be w placed at positions $0, \dots, |w| - 1$ and 1 everywhere else. Every 0 of x lies inside w . So a block $01^{2k+1}0$ of x would lie inside w , which is impossible. Hence $x \in X$ and $w \in \mathcal{L}(X)$. $\square$

4 The Fischer cover has two states

Let G be the graph with vertices $A = 0$, $B = 1$ and edges

$$A \xrightarrow{0} A, \quad A \xrightarrow{1} B, \quad B \xrightarrow{1} A.$$

It is right-resolving and irreducible. It is follower-separated: 0 labels a path from A but not from B .

Proposition 2. $X_G = X$.

Proof. ($X_G \subseteq X$.) Let (s_i) be a path with labels (x_i) . Suppose $01^{2k+1}0$ occurs at i . The only 0-edge ends at A , so $s_{i+1} = A$. Along the 1-edges the state alternates, so $s_{i+1+t} = A$ for even t and B for odd t ($t \leq 2k+1$). Hence $s_{i+2k+2} = B$. But $x_{i+2k+2} = 0$ and B has no 0-edge. This is a contradiction.

($X \subseteq X_G$.) Let $x \in X$. We say position i is in state B if one of the following holds:

- (a) there is a 0 at or after i , and the first one is at $i+t$ with t odd;
- (b) there is no 0 at or after i , there is a 0 before i , and the last one is at $i-1-t$ with t odd;
- (c) x has no 0 at all and i is odd.

Otherwise i is in state A . We check that $s_i \xrightarrow{x_i} s_{i+1}$ is an edge for every i .

If $x_i = 0$, then (a) fails at i with $t = 0$, and (b), (c) fail too. So $s_i = A$. At $i+1$, (a) would give a block 01^t0 at i with t odd, which is excluded since $x \in X$. (b) fails because the last 0 before $i+1$ is at i , so $t = 0$. (c) fails. So $s_{i+1} = A$ and $A \xrightarrow{0} A$ is an edge.

If $x_i = 1$, we show that s_{i+1} is the other state from s_i . Then the step is $A \xrightarrow{1} B$ or $B \xrightarrow{1} A$. If there is a 0 after i , say first at $i+1+t$, then the first 0 at or after i is at $i+(t+1)$. Only rule (a) applies at both positions, with parities of t and $t+1$. If there is no 0 after i but one before i , say last at $i-1-t$, then the last 0 before $i+1$ is at $(i+1)-1-(t+1)$. Only rule (b) applies, with parities of t and $t+1$. If x has no 0 at all, rule (c) applies with the parities of i and $i+1$. $\square$

Proposition 3. No labelled graph with at most one vertex presents X .

Proof. A graph with no vertex has no bi-infinite path, but $0^\infty \in X$. A graph with one vertex and loop label set $T \subseteq \{0, 1\}$ presents the full shift $T^\mathbb{Z}$. If $0 \notin T$, it misses $0^\infty \in X$. If $1 \notin T$, it misses $1^\infty \in X$ (which has no 0). If $T = \{0, 1\}$, it contains $\cdots 0001000 \cdots \notin X$, since this point contains the block 010. $\square$

Theorem 4. $\text{Fis}(X) = \text{Pres}(X) = 2$. The graph G is the Fischer cover of X .

Proof. By Propositions 2 and 3. The Fischer cover is the unique minimal right-resolving presentation, so it is G . $\square$

The verification script also enumerates all 2^8 labelled graphs on two vertices. Exactly two of them have the same blocks as X up to length 8, and these are G and its relabelling.

5 The Hankel matrix has rank 3

By Lemma 1, $H[u][v] = 1$ iff uv has no factor $01^{\text{odd}}0$. Take rows $u \in \{11, 0, 01\}$ and columns $v \in \{0, 10, 1\}$:

$$\begin{array}{c|ccc} & 0 & 10 & 1 \\ \hline 11 & 110 \in \mathcal{L} & 1110 \in \mathcal{L} & 111 \in \mathcal{L} \\ 0 & 00 \in \mathcal{L} & 010 \notin \mathcal{L} & 01 \in \mathcal{L} \\ 01 & 010 \notin \mathcal{L} & 0110 \in \mathcal{L} & 011 \in \mathcal{L} \end{array} \quad M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

We have $\det M = 1 \cdot (0 - 1) - 1 \cdot (1 - 0) + 1 \cdot (1 - 0) = -1$. The integer matrix $N = \begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$ satisfies $MN = NM = I$. So M is invertible over every commutative ring, and in particular over every field.

Theorem 5. *Over every field K , the Hankel matrix of $\mathcal{L}(X)$ has rank exactly 3.*

Proof. The rank is at least 3 because M is a nonsingular submatrix. For the upper bound we show that H has at most four distinct rows, one of which is zero. If $u \notin \mathcal{L}(X)$, then $\mathcal{F}(u) = \emptyset$, because $\mathcal{L}(X)$ is closed under taking factors. Let $u \in \mathcal{L}(X)$. A forbidden block in uv lies inside u (impossible), inside v , or straddles the junction. A straddling block 01^m0 has its first 0 in u and all 1s up to the junction, so it starts at the last 0 of u . Write $u = u'01^j$ if u contains a 0, and $v = 1^r0v'$ if v contains a 0. Then $uv \in \mathcal{L}(X)$ iff $v \in \mathcal{L}(X)$ and, when both contain a 0, $j + r$ is even. Hence $\mathcal{F}(u) = \mathcal{L}(X)$ if u has no 0, $\mathcal{F}(u) = \mathcal{F}(0)$ if j is even, and $\mathcal{F}(u) = \mathcal{F}(01)$ if j is odd. The rows lie in the span of the three vectors $\mathbf{1}_{\mathcal{L}(X)}$, $\mathbf{1}_{\mathcal{F}(0)}$ and $\mathbf{1}_{\mathcal{F}(01)}$, so the rank is at most 3. $\square$

Corollary 6. $\text{Fis}(X) = 2 \neq 3 = \text{rank}_K H$ for every field K . Clause (C), and with it Conjecture 00000009617, is false.

6 A second counterexample: the charge-constrained shift

The even shift has exactly 3 nonempty follower sets of words, and this number equals its Hankel rank. So the even shift alone does not refute the reading in which “minimal state count” means the size of the follower-set graph. The following shift separates all of these numbers.

Let G_c be the graph with vertices 0, 1, 2 and edges

$$0 \xrightarrow{1} 1, \quad 1 \xrightarrow{1} 2, \quad 2 \xrightarrow{0} 1, \quad 1 \xrightarrow{0} 0,$$

and let $X_c = X_{G_c}$ be the shift it presents. Read 1 as charge +1 and 0 as charge −1. A word is a path label iff its running charge (the partial sums) stays in a band of width 2. So X_c is the charge-constrained shift with 3 charge levels [2, Ex. 1.2.7]. G_c is irreducible and right-resolving. It is follower-separated:

- the word 0 labels a path from vertices 1 and 2 but not from 0;
- the word 1 labels a path from 0 and 1 but not from 2;
- the word 11 labels a path from 0 but not from 1.

Proposition 7. 1. $\text{Fis}(X_c) = 3$. Moreover, no presentation of X_c has fewer than 3 vertices.

2. The terminal vertex sets $T(u) = \{q : \text{some path labelled } u \text{ ends at } q\}$ of the words $u \in \mathcal{L}(X_c)$ are $\{0, 1, 2\}$, $\{0, 1\}$, $\{1, 2\}$, $\{0\}$, $\{2\}$, $\{1\}$. They are attained by $u = \varepsilon, 0, 1, 00, 11, 001$. X_c has exactly 6 distinct nonempty follower sets $\mathcal{F}(u)$, and 7 with $\emptyset$.

3. Over every field K , the Hankel matrix H_c of $\mathcal{L}(X_c)$ has rank exactly 5.

Proof. (1) G_c is an irreducible, right-resolving, follower-separated presentation, so it is the Fischer cover [2, §3.3]. For arbitrary presentations, `verify.py` checks all $1 + 4 + 256$ labelled graphs with at most 2 vertices. Each one has a different set of blocks of length ≤ 8 .

(2) A right-resolving graph has $\mathcal{F}(u) = \bigcup_{q \in T(u)} \mathcal{F}_{G_c}(q)$, where $\mathcal{F}_{G_c}(q)$ is the set of labels of paths starting at q . Reading 0 moves every vertex down by one, and reading 1 moves every vertex up by one, inside $\{0, 1, 2\}$. So $T(u)$ is always an interval. All six intervals occur, as listed. Distinct intervals give distinct follower sets, by follower separation and the words 0, 1, 11.

(3) The lower bound: rows and columns $\varepsilon, 0, 1, 00, 11$ give the minor

$$M_c = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad \det M_c = 1.$$

Its entries are 0 exactly for the words 000, 0000, 111, 1111. A determinant equal to 1 is a unit in every ring.

The upper bound: write $A = \mathcal{F}_{G_c}(0)$, $B = \mathcal{F}_{G_c}(1)$, $C = \mathcal{F}_{G_c}(2)$. A nonempty word in A starts with 1, and a nonempty word in C starts with 0. So $A \cap C = \{\varepsilon\} \subseteq B$. Hence

$$\mathbf{1}_{A \cup B} + \mathbf{1}_{B \cup C} = \mathbf{1}_{A \cup B \cup C} + \mathbf{1}_{B \cup (A \cap C)} = \mathbf{1}_{A \cup B \cup C} + \mathbf{1}_B,$$

that is, $\mathbf{1}_{\mathcal{F}(0)} + \mathbf{1}_{\mathcal{F}(1)} = \mathbf{1}_{\mathcal{F}(\varepsilon)} + \mathbf{1}_{\mathcal{F}(001)}$. By (2), every row of H_c is zero or one of the six follower sets. Those six span a space of dimension at most 5. So $\text{rank}_K H_c \leq 5$. $\square$

Corollary 8. For X_c the Hankel rank is 5, which differs from:

- the Fischer count 3;
- the least size 3 of any presentation;
- the number 6 of nonempty follower sets (the follower-set graph, which is the trim minimal deterministic automaton);
- the size 7 of the complete minimal deterministic automaton.

7 Readings of the clause

Only the standard reading, rank over a field, is refuted. Under that reading, the counterexamples work for every interpretation of “minimal state count” listed below.

- **Rank over a field** ($\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{F}_p$, or any other field). For the even shift the rank is $3 \neq 2$, and for X_c it is $5 \neq 3$. The minors have determinant ± 1 , so neither counterexample depends on the field. This is the standard meaning of the rank of a Hankel matrix (Section 2).
- **Which state count.** Possible meanings, with their values:
 - the minimal right-resolving presentation: 2 for X , 3 for X_c ;
 - the least number of states of *any* presentation: 2 for X , 3 for X_c ;

- the left Fischer cover: 2 for X , since X is closed under reversal and Proposition 3 holds for all graphs; 3 for X_c , since X_c is also closed under reversal (the running charge of a reversed word has the same range);
- the number of nonempty follower sets (the follower-set graph or trim minimal automaton): 3 for X , which equals its rank, but $6 \neq 5$ for X_c ;
- the complete minimal automaton: $4 \neq 3$ for X and $7 \neq 5$ for X_c .
- **Indexing of H .** Rows and columns may range over all words, over nonempty words, or over words of length at most n for any $n \geq 2$. All of these contain M , so the rank of the even shift's matrix is at least 3 in each case. For X_c , indexing sets that contain $\varepsilon, 0, 1, 00, 11$ contain M_c . Indexing by $u^R v$ or transposing does not change the rank.
- **Boolean rank (not refuted).** Over the Boolean semiring (OR/AND), the rank of the even shift's H is 2. Indeed $\mathbf{1}_{\mathcal{L}(X)} = \mathbf{1}_{\mathcal{F}(0)} \vee \mathbf{1}_{\mathcal{F}(01)}$: if v has no 0 then $v \in \mathcal{F}(0)$; otherwise $v \in \mathcal{F}(0)$ or $v \in \mathcal{F}(01)$ by the parity of its leading run of 1s. The fooling set $\{(0,0), (01,10)\}$ shows that the Boolean rank is at least 2.

More generally, the Boolean rank is *always* at most the Fischer count. If G is a right-resolving presentation with n vertices, then $\mathcal{F}(u) = \bigcup_{q \in T(u)} \mathcal{F}_G(q)$. So H is the Boolean product of the $\{\text{words}\} \times n$ matrix $[q \in T(u)]$ and the $n \times \{\text{words}\}$ matrix $[v \in \mathcal{F}_G(q)]$. Searches over small examples found no case where the two differ, so the clause may well be *true* under the Boolean reading. We refute only the field-rank reading. Boolean rank is a different notion and is always named as such. Over $\mathbb{Q}$, $\mathbf{1}_{\mathcal{L}(X)} \neq \mathbf{1}_{\mathcal{F}(0)} + \mathbf{1}_{\mathcal{F}(01)}$, because both follower sets contain every word 1^k .

- **Hidden Markov model reading.** One could instead take the Hankel matrix of the probability function $P(w)$ of a measure on the shift. For a Markov measure carried by the Fischer cover (a hidden Markov model with n hidden states), this rank is at most the Fischer count n . For the even shift with the Parry measure, or with any fully supported measure of this kind, it is exactly 2. `verify.py` computes 2 for a rational Markov measure. But the value depends on the measure: the point mass at $0^\infty \in X$ gives rank 1. So it is not an invariant of the shift. The clause speaks of the Hankel matrix of the *language*, i.e. of the indicator function of $\mathcal{L}(X)$, and that is what we use.

8 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project. It uses only the core library (no Mathlib). Points are functions `Int → Bool` (`false` is 0, `true` is 1).

- `EvenShift x`: no `ForbiddenAt x i k`, the block $01^{2k+1}0$ at i . `Lang X w`: some point of X contains w .
- `LGraph n, IsPathLabel, Presents E X` (X equals the set of bi-infinite path labels), `RightResolving, Sofic`. `IsFisherCount X m` means that m is the least number of vertices of a right-resolving presentation. `IsMinPresCount` is the same for arbitrary presentations.
- `hankelEntry X u v` is 1 if `Lang X (u ++ v)` holds and 0 otherwise. `minor`, and `det` (Laplace expansion). `HankelRankAtLeast X r` means that some $r \times r$ minor has nonzero integer determinant, i.e. the rank over $\mathbb{Q}$ is at least r . `HankelRankAtLeastMod p` means that the determinant is nonzero mod p . We prove it for every $p \geq 2$, in particular for every

prime p , where it means rank over $\mathbb{F}_p$. For composite p , $\mathbb{Z}/p$ is not a field, but our minors are unimodular. `HankelRankIs X m` means at least m and not at least $m + 1$.

- **Claim:** for every sofic X and every m with `IsFisherCount X m`, `HankelRankIs X m`. `ClaimMod p` and `ClaimAnyPresentation` are the variants for $\mathbb{Z}/p$ and for arbitrary presentations.
- Even shift, presentation:
 - `lang_iff`: Lemma 1, for all words.
 - `fisher_sound`, `fisher_complete`, `fisher_presents`: Proposition 2.
 - `no_zero_state`, `no_one_state`: Proposition 3.
 - `fisher_count`, `min_presentation_count`: Theorem 4.
 - `fisher_rightResolving`, `fisher_irreducible_separated`: the properties of G .
- Even shift, Hankel rank: `minor_eq`, `minor_det` ($= -1$), `minor_unimodular` (the inverse N), `rows_independent` (the three rows are linearly independent over $\mathbb{Q}$), `hankel_rank_ge_three` and `hankel_rank_ge_three_mod` (for every $p \geq 2$).
- Even shift, main theorems: `conjecture_00000009617_false` : $\neg \text{Claim}$;
`conjecture_00000009617_false_mod` ($\forall p \geq 2, \neg \text{ClaimMod } p$);
`conjecture_00000009617_false_any_presentation`. `fisher_lt_rank`: every m with `IsFisherCount X m` equals 2, and the rank is at least $m + 1$.
- X_c :
 - `charge` is G_c , and `ChargeShift` is the shift it presents.
 - `charge_rightResolving`, `charge_irreducible_separated`: the properties of G_c .
 - `not_lang_of_bad`: 000 and 111 do not occur in X_c .
 - `per_mem` and `lang_of_per`: the periodic point $(1100)^\infty$ lies in X_c , and all its factors are in $\mathcal{L}(X_c)$.
 - `chargeMinor_eq`, `chargeMinor_det` ($= 1$): the minor M_c .
 - `charge_minors`: minors with determinant ± 1 of every size $r \leq 5$.
 - `charge_rank_ge`, `charge_rank_ge_mod`: rank at least r for every $r \leq 5$.
 - `charge_counterexample`: the Fischer count exists and is ≤ 3 (`exists_min`), and the rank is ≥ 5 .
 - `conjecture_00000009617_false_charge`, `_charge_mod`, `_charge_any_presentation`: three more proofs of $\neg \text{Claim}$, $\neg \text{ClaimMod } p$ and $\neg \text{ClaimAnyPresentation}$.

Non-vacuity: `evenShift_sofic`, `fisher_count` and `charge_counterexample` show that the hypotheses of `Claim` hold. `allZero_mem`, `allOne_mem` and `single_one_not_mem` show that X is neither empty nor full. The final examples show that `Lang` is nontrivial and that `det` returns 0 on a singular matrix.

Limitations. Lean proves rank *at least* 3 for the even shift and at least 5 for X_c , which suffices. For X_c it proves Fischer count at most 3, which also suffices. The following are proved in this report and checked on finite truncations by `verify.py`, but not in Lean:

- the upper bounds on the rank;
- $\text{Fis}(X_c) = 3$ exactly;
- the count of 6 follower sets, so the follower-set reading is not in Lean;
- the Boolean-rank and measure remarks.

Lean works over the alphabet $\{0, 1\}$. The clause quantifies over all sofic shifts, so binary counterexamples suffice. The project has no `sorry`, no `native_decide` and no added axioms. `#print axioms` reports only the standard `propext`, `Classical.choice` and `Quot.sound`. Classical logic is used to define the path in `fisher_complete`, the Hankel entries of an arbitrary shift, and the minimum in `exists_min`.

9 Independent check

`verify.py` (Python 3 standard library) checks the following.

- For the even shift:
 - three descriptions of $\mathcal{L}(X)$ agree on all words of length at most 14 (forbidden blocks, parities of inner runs, paths in G);
 - among all labelled graphs with at most 2 vertices, none with fewer than 2 vertices matches, and the matching 2-vertex graphs are G up to relabelling;
 - the minor, its determinant -1 and its inverse;
 - the Hankel truncation on all 127 words of length at most 6 has rank 3 over $\mathbb{Q}$ and over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$;
 - only 4 distinct rows occur;
 - the Boolean-rank identity and the Boolean factorization through the Fischer vertices hold;
 - the measure Hankel ranks are 2 for a Markov measure and 1 for the point mass.
- For X_c :
 - graph paths agree with the charge-band description up to length 14;
 - no graph with at most 2 vertices matches;
 - the six terminal sets, and 6 nonzero rows (7 with zero);
 - M_c with $\det M_c = 1$, and the row relation;
 - rank 5 of the truncation over $\mathbb{Q}$ and over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$.

10 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # standard library only
pdflatex report.tex && pdflatex report.tex
```

References

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